

Machine Learning

Lecture 9: Theory of Generalization (III): *Variance-Bias Decomposition*

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本节内容

- Variance-Bias decomposition
- Learning curves

Approximation-generalization tradeoff

- Small E_{out} : good approximation of f out of sample
- More complex $\mathcal{H} \implies$ better chance of approximating f
- Less complex $\mathcal{H} \implies$ better chance of generalizing out of sample
- Ideal $\mathcal{H} = \{f\}$
- VC analysis was one approach: $E_{out} \leq E_{in} + \Omega$
- Bias-variance analysis is another: decomposing E_{out} into
 - A. How well $\mathcal{H}$ can approximate f
 - B. How well we can zoom in on a good $h \in \mathcal{H}$
- Applies to real-valued targets and uses squared error

Decomposing E_{out}

$$E_{out}(g^{(\mathcal{D})}) = \mathbb{E}_{\mathbf{x}} \left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right]$$

$$\begin{aligned} \mathbb{E}_{\mathcal{D}} \left[E_{out}(g^{(\mathcal{D})}) \right] &= \mathbb{E}_{\mathcal{D}} \left[\mathbb{E}_{\mathbf{x}} \left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right] \right] \\ &= \mathbb{E}_{\mathbf{x}} \left[\mathbb{E}_{\mathcal{D}} \left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right] \right] \end{aligned}$$

Now, let us focus on:

$$\mathbb{E}_{\mathcal{D}} \left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right]$$

The average hypothesis

To evaluate $\mathbb{E}_{\mathcal{D}} \left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right]$

we define the ‘average’ hypothesis $\bar{g}(\mathbf{x})$:

$$\bar{g}(\mathbf{x}) = \mathbb{E}_{\mathcal{D}} \left[g^{(\mathcal{D})}(\mathbf{x}) \right]$$

Imagine **many** data sets $\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_K$

$$\bar{g}(\mathbf{x}) \approx \frac{1}{K} \sum_{k=1}^K g^{(\mathcal{D}_k)}(\mathbf{x})$$

Using the average hypothesis

$$\begin{aligned}\mathbb{E}_{\mathcal{D}} \left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right] &= \mathbb{E}_{\mathcal{D}} \left[\left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}) + \bar{g}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right] \\&= \mathbb{E}_{\mathcal{D}} \left[\left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}) \right)^2 + \left(\bar{g}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right. \\&\quad \left. + 2 \left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}) \right) \left(\bar{g}(\mathbf{x}) - f(\mathbf{x}) \right) \right] \\&= \mathbb{E}_{\mathcal{D}} \left[\left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}) \right)^2 \right] + \left(\bar{g}(\mathbf{x}) - f(\mathbf{x}) \right)^2\end{aligned}$$

Bias and Variance

$$\mathbb{E}_{\mathcal{D}} \left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right] = \underbrace{\mathbb{E}_{\mathcal{D}} \left[\left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}) \right)^2 \right]}_{\text{var}(\mathbf{x})} + \underbrace{\left(\bar{g}(\mathbf{x}) - f(\mathbf{x}) \right)^2}_{\text{bias}(\mathbf{x})}$$

$$\text{Therefore, } \mathbb{E}_{\mathcal{D}} \left[E_{\text{out}}(g^{(\mathcal{D})}) \right] = \mathbb{E}_{\mathbf{x}} \left[\mathbb{E}_{\mathcal{D}} \left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right] \right]$$

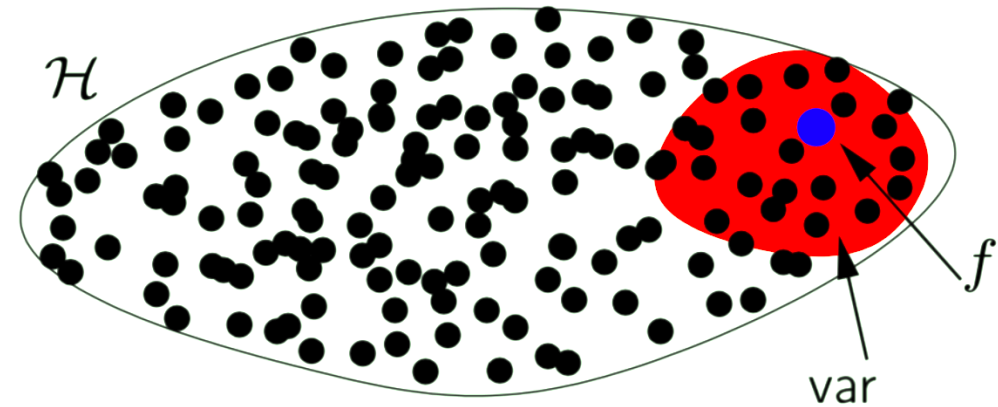
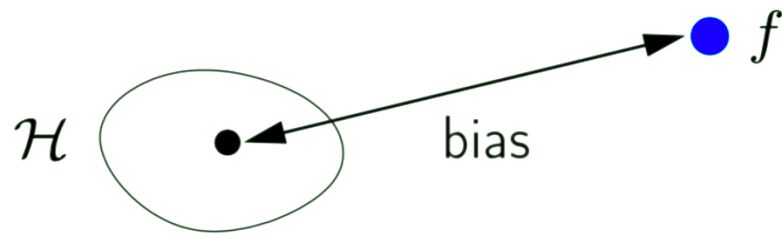
$$= \mathbb{E}_{\mathbf{x}} [\text{bias}(\mathbf{x}) + \text{var}(\mathbf{x})]$$

$$= \text{bias} + \text{var}$$

The tradeoff

$$\text{bias} = \mathbb{E}_{\mathbf{x}} \left[\left(\bar{g}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right]$$

$$\text{var} = \mathbb{E}_{\mathbf{x}} \left[\mathbb{E}_{\mathcal{D}} \left[\left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}) \right)^2 \right] \right]$$



$\mathcal{H} \uparrow$



Example: sin target

$$f : [-1, 1] \rightarrow \mathbb{R} \quad f(x) = \sin(\pi x)$$

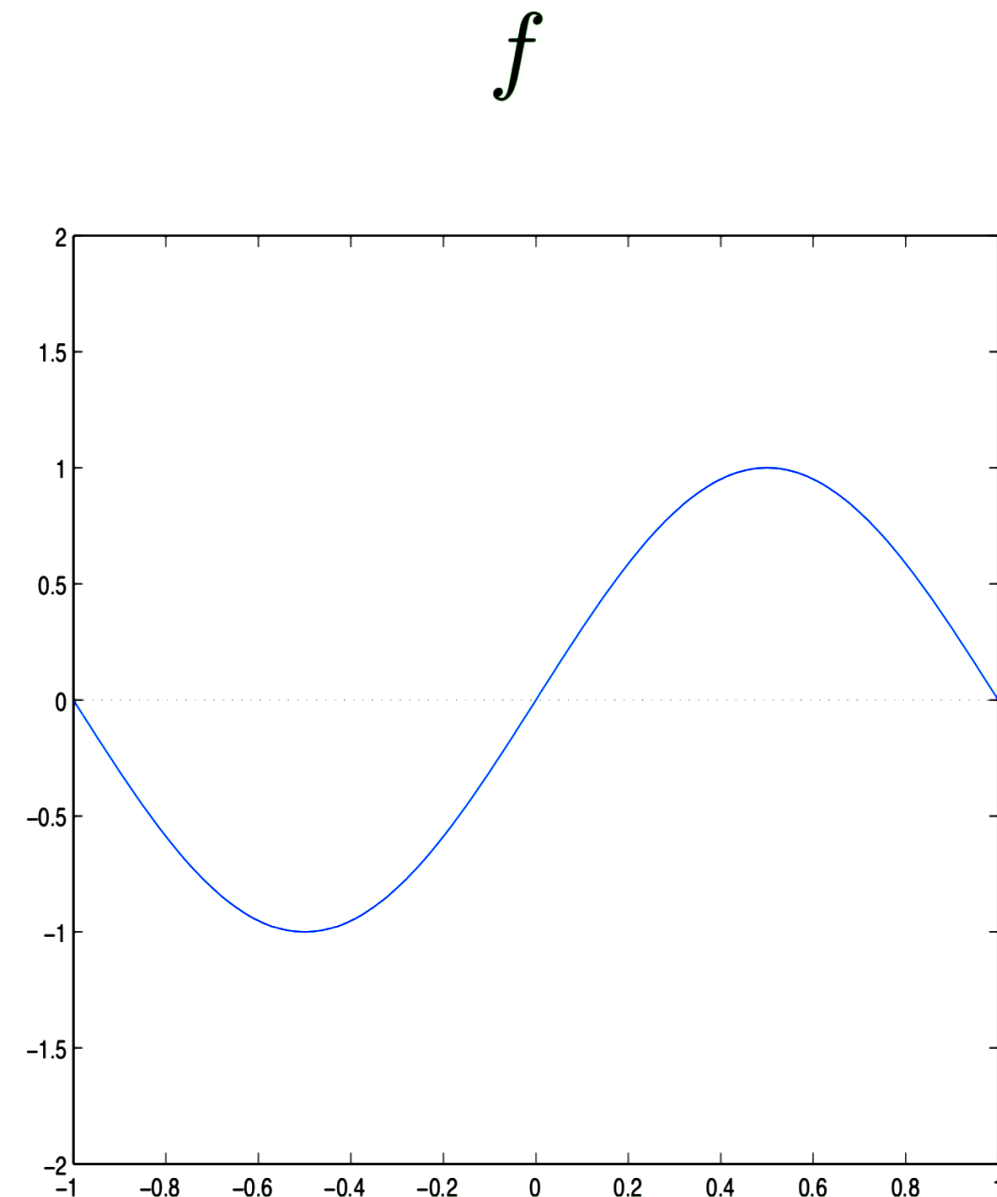
Only two training examples! $N = 2$

Two models used for learning:

$$\mathcal{H}_0: \quad h(x) = b$$

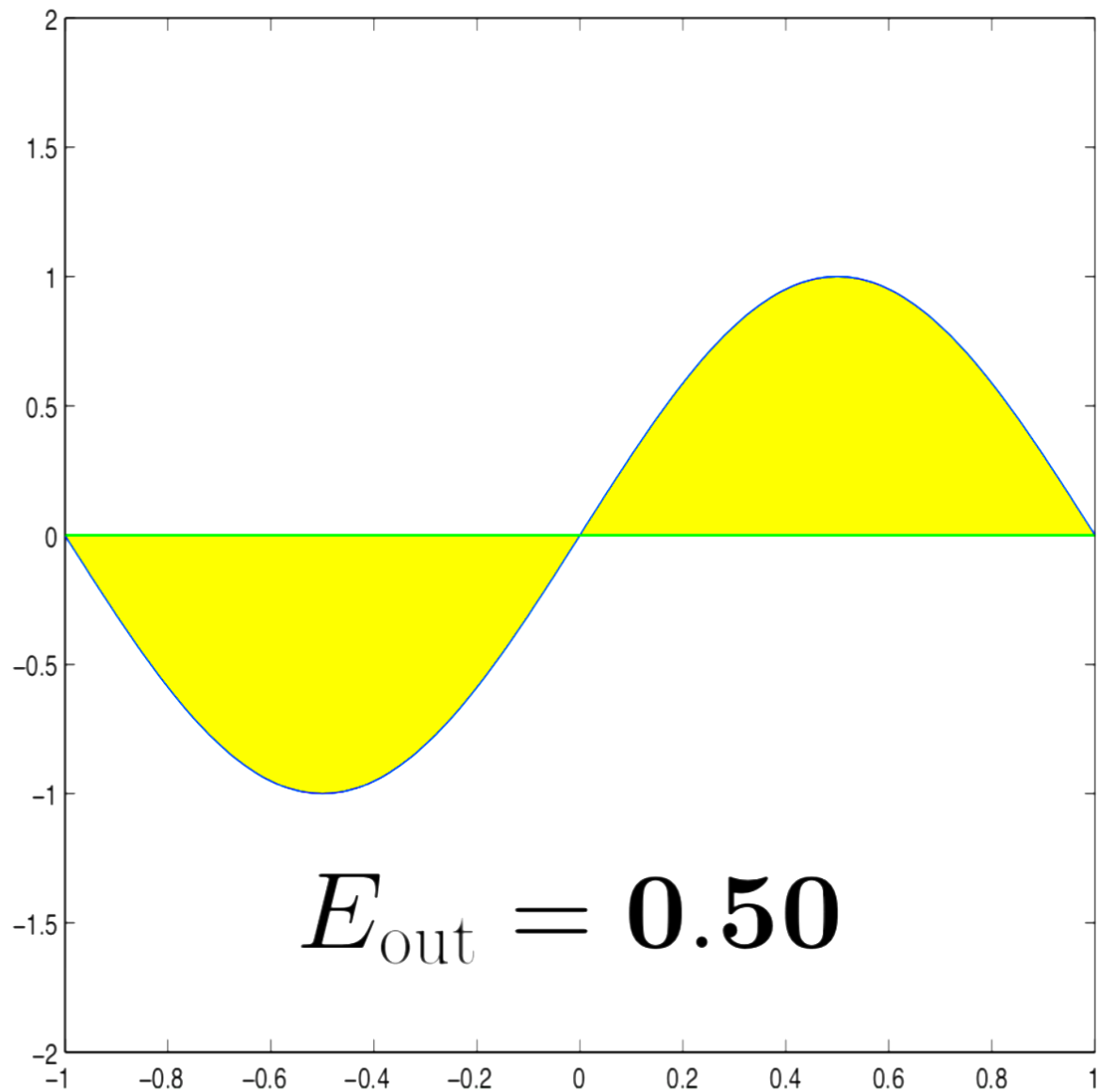
$$\mathcal{H}_1: \quad h(x) = ax + b$$

Which is better, $\mathcal{H}_0$ or $\mathcal{H}_1$?

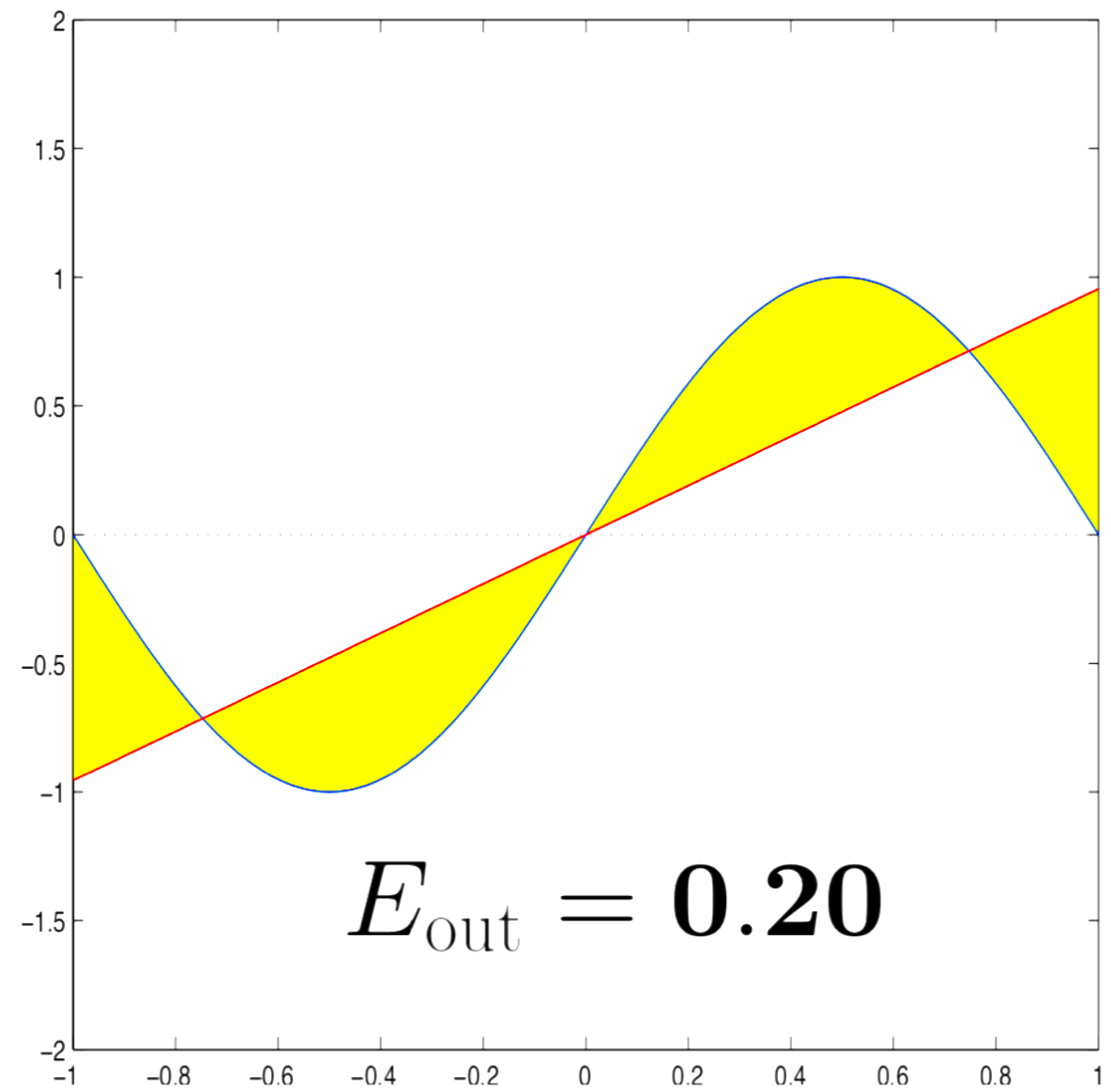


Approximation: $\mathcal{H}_0$ versus $\mathcal{H}_1$

$\mathcal{H}_0$

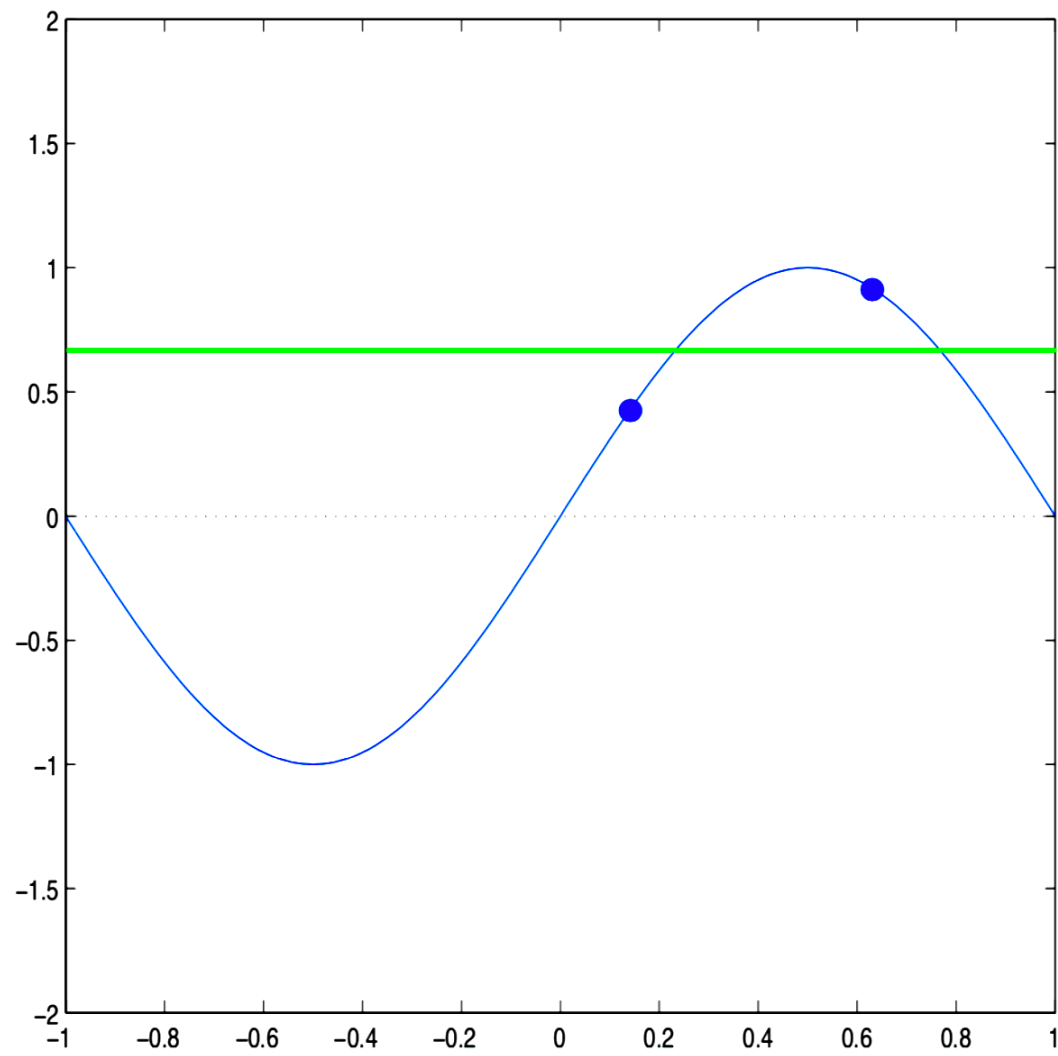


$\mathcal{H}_1$

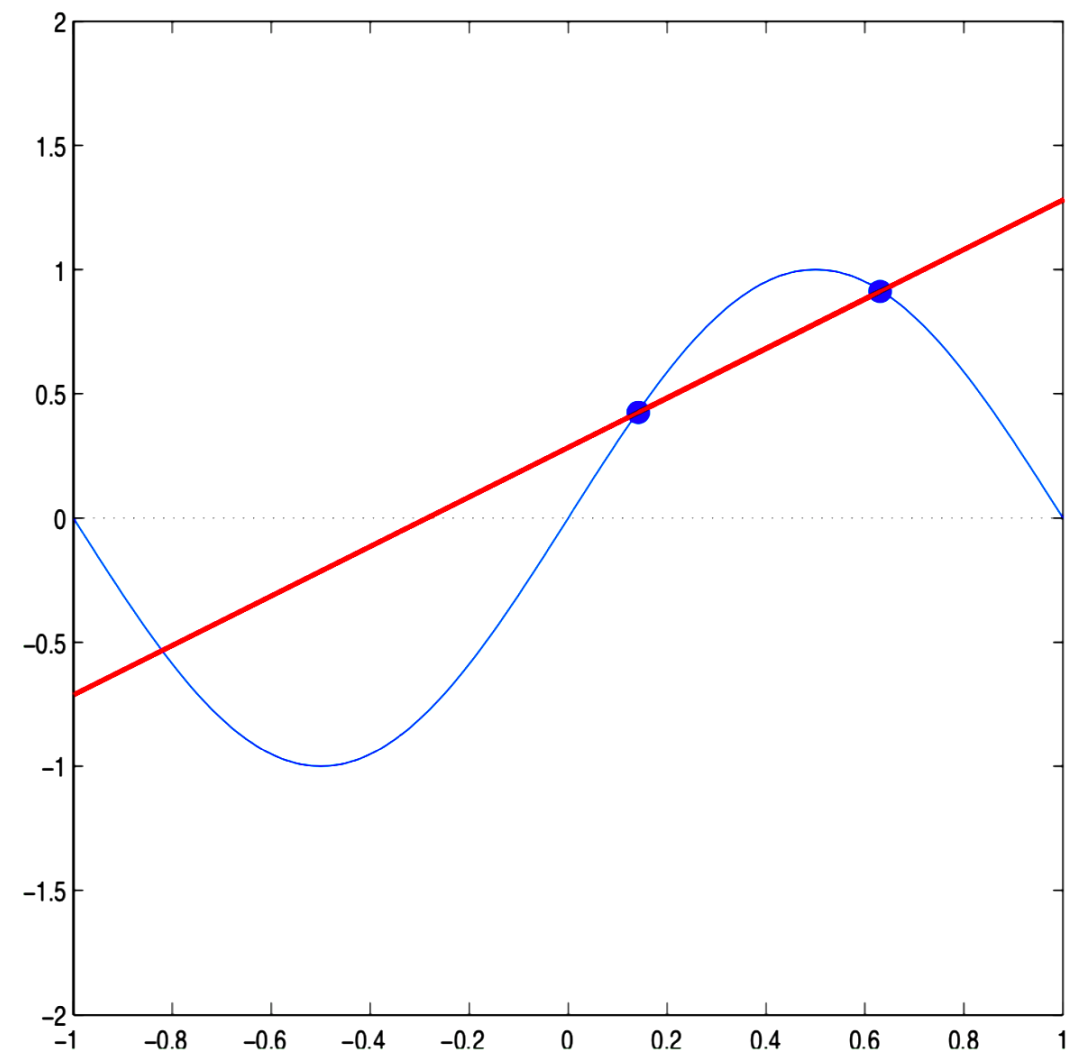


Learning: $\mathcal{H}_0$ versus $\mathcal{H}_1$

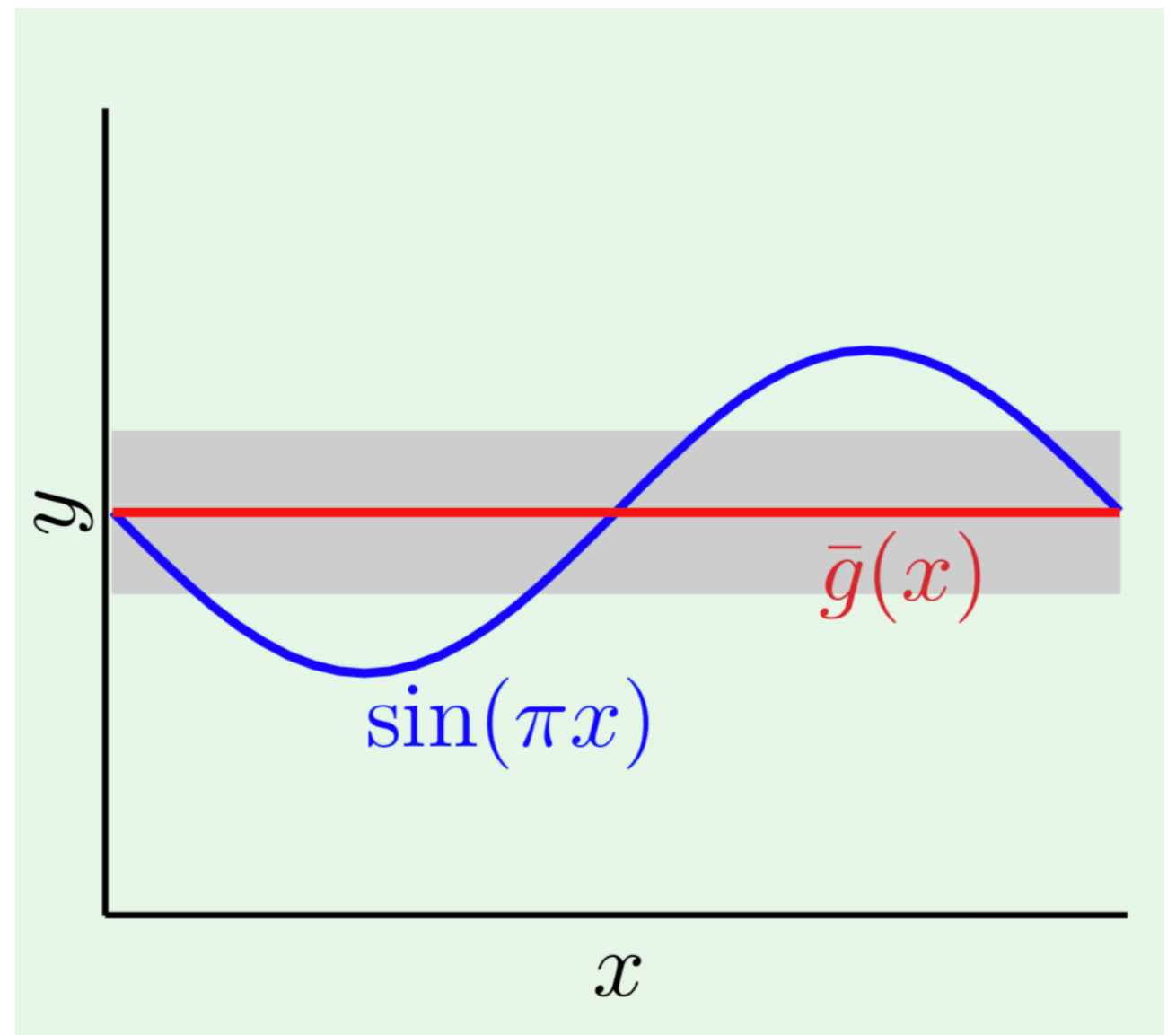
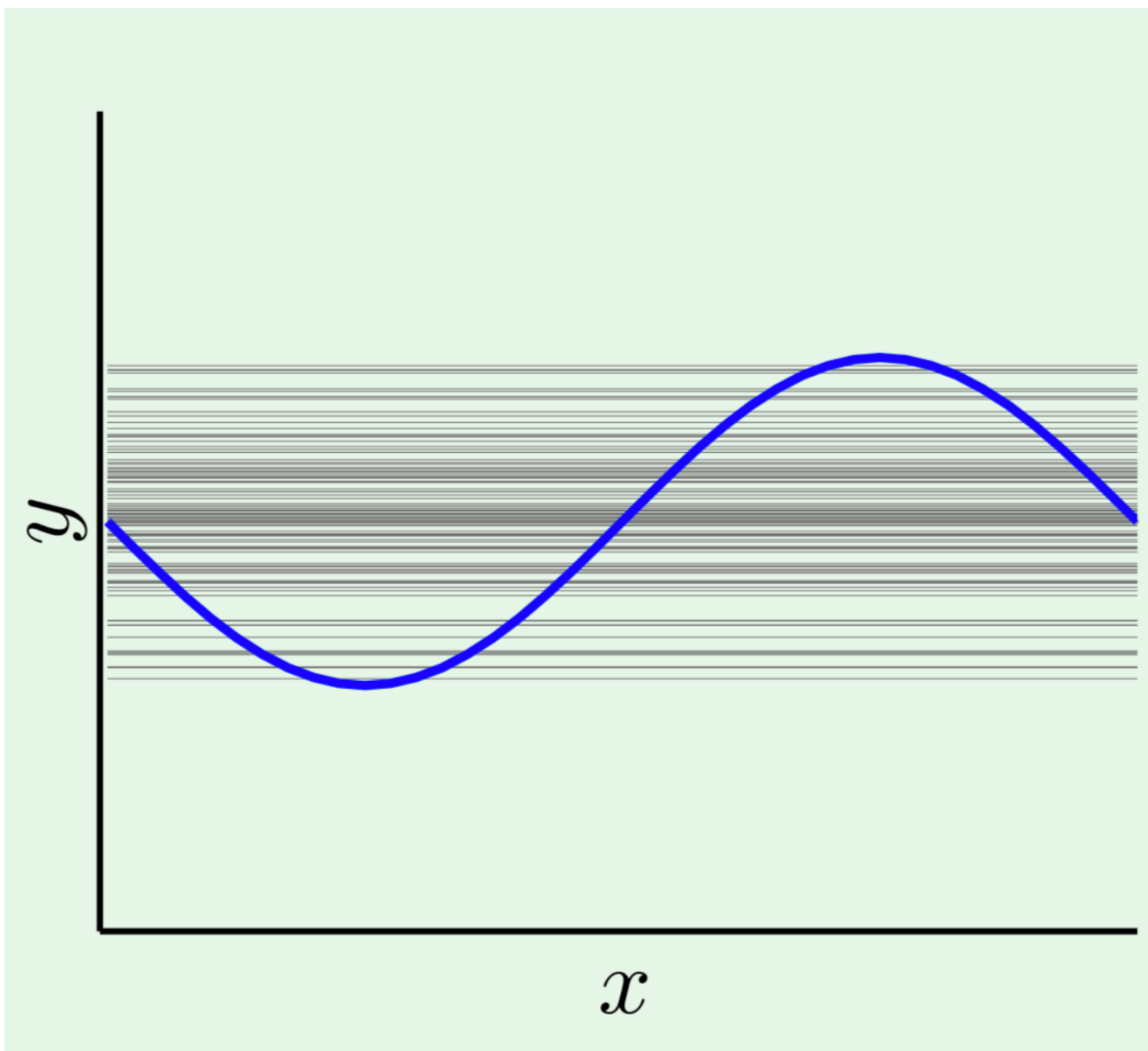
$\mathcal{H}_0$



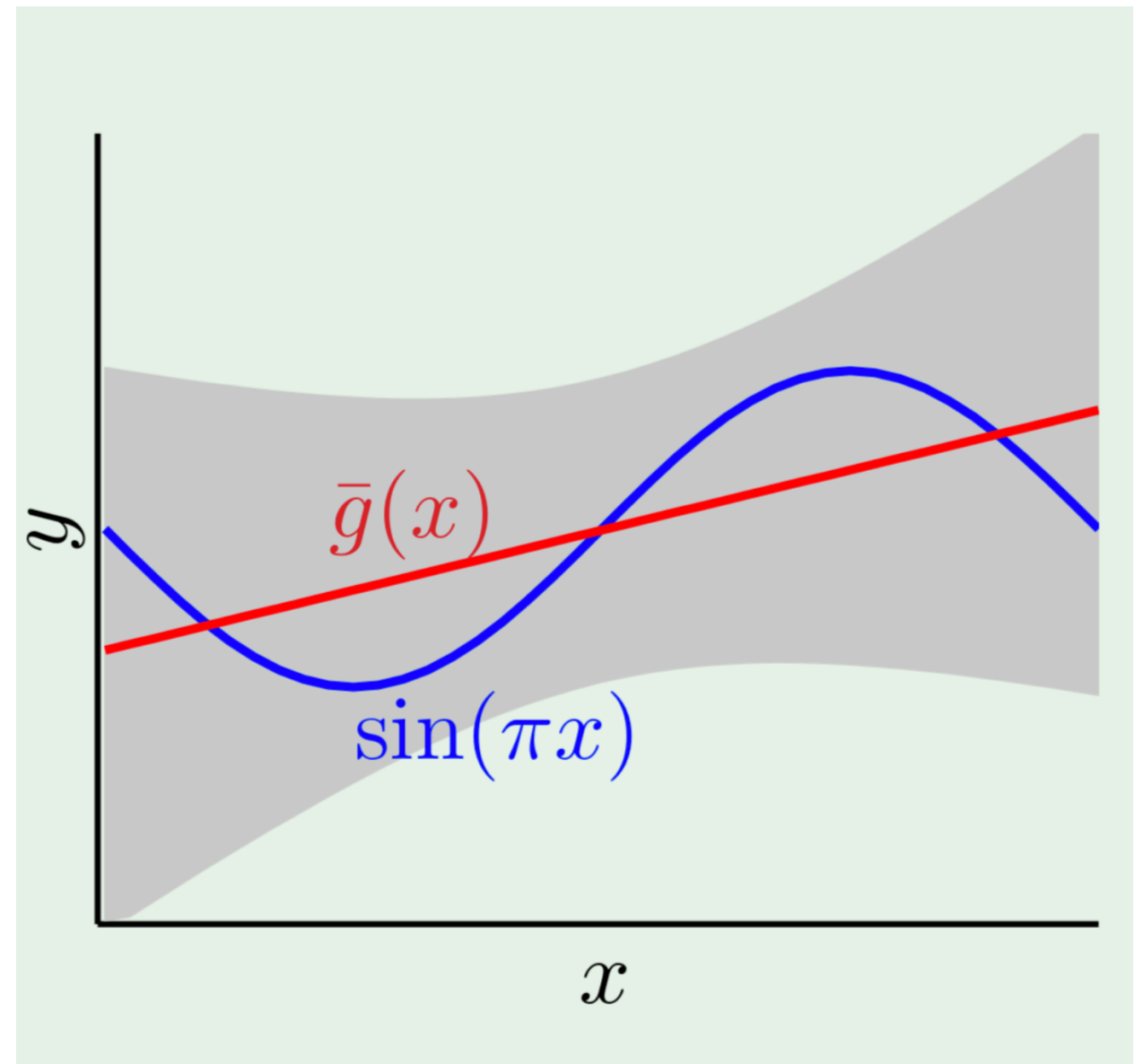
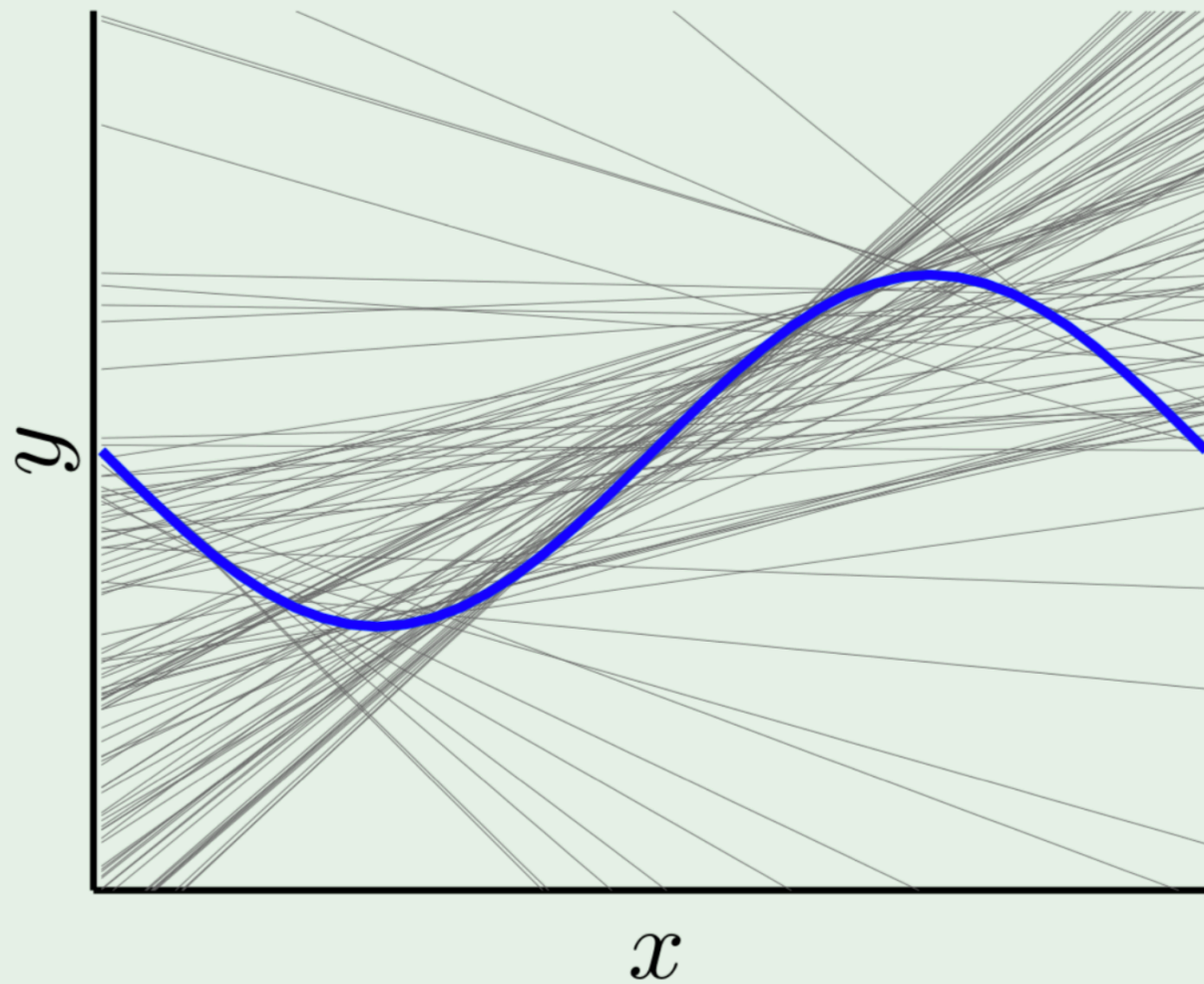
$\mathcal{H}_1$



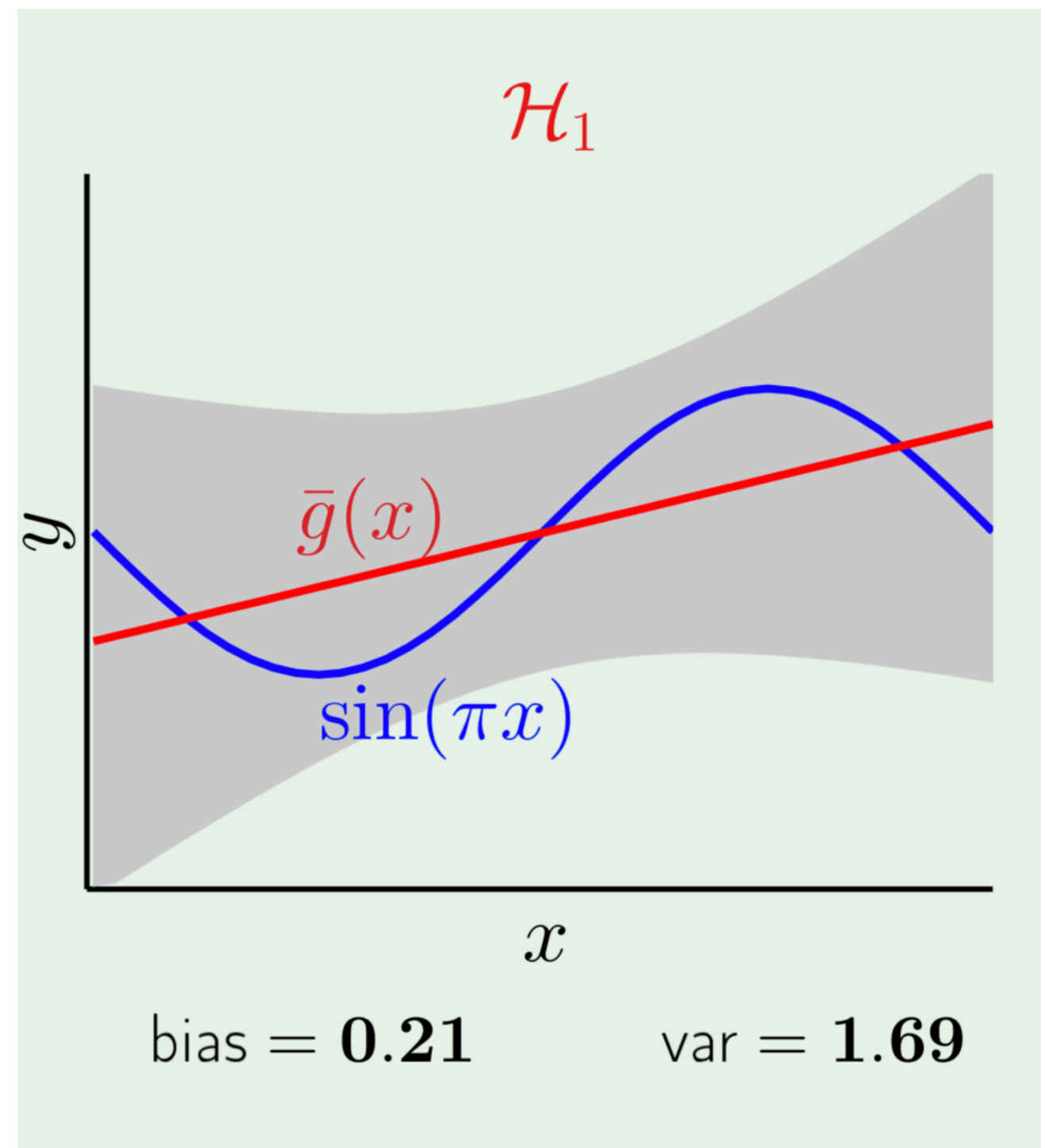
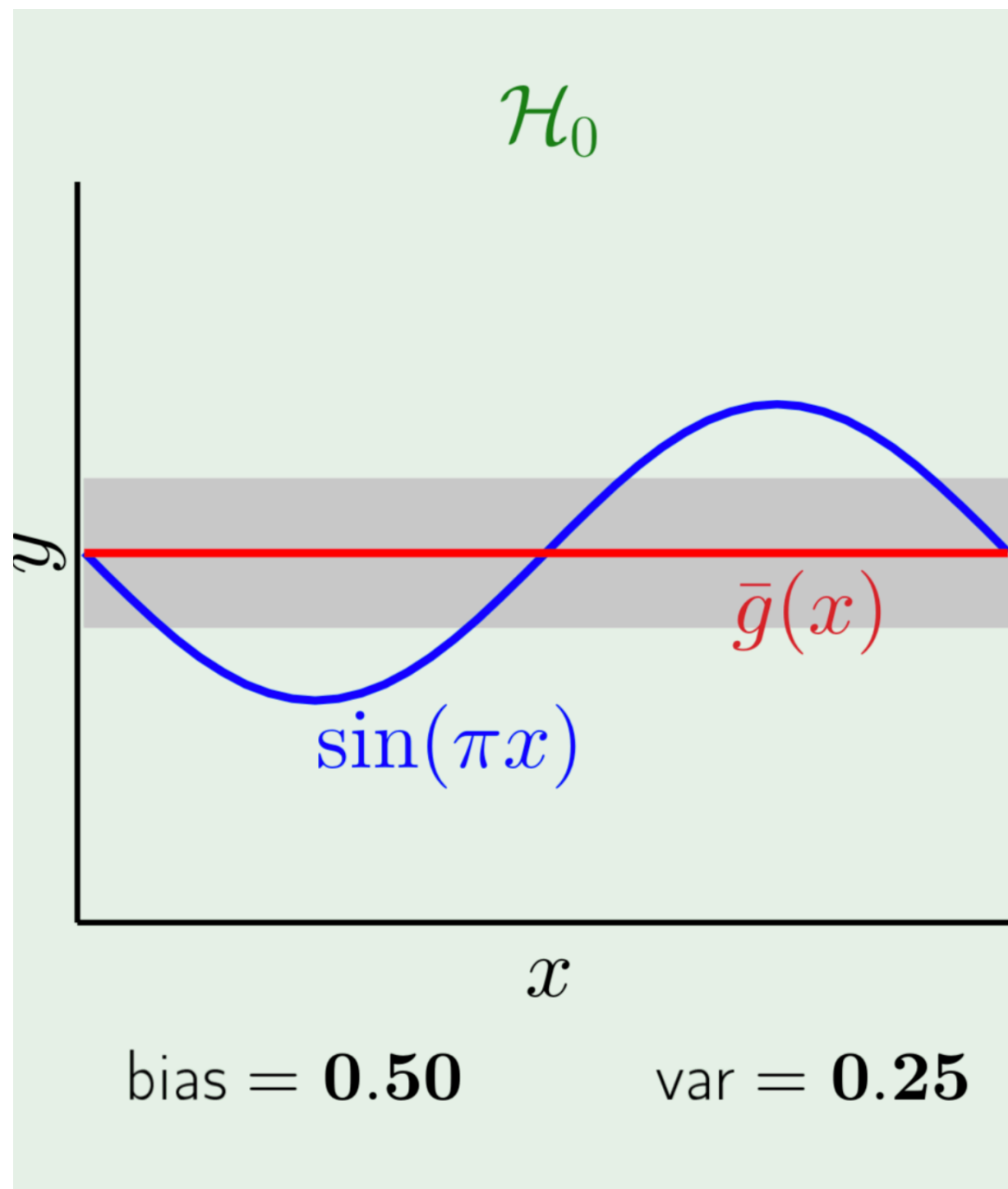
Bias and variance: $\mathcal{H}_0$



Bias and variance: $\mathcal{H}_1$



The winner is...



Conclusion:

**Match the “model complexity” to the data resource,
NOT to the target complexity**

Expected E_{out} and E_{in}

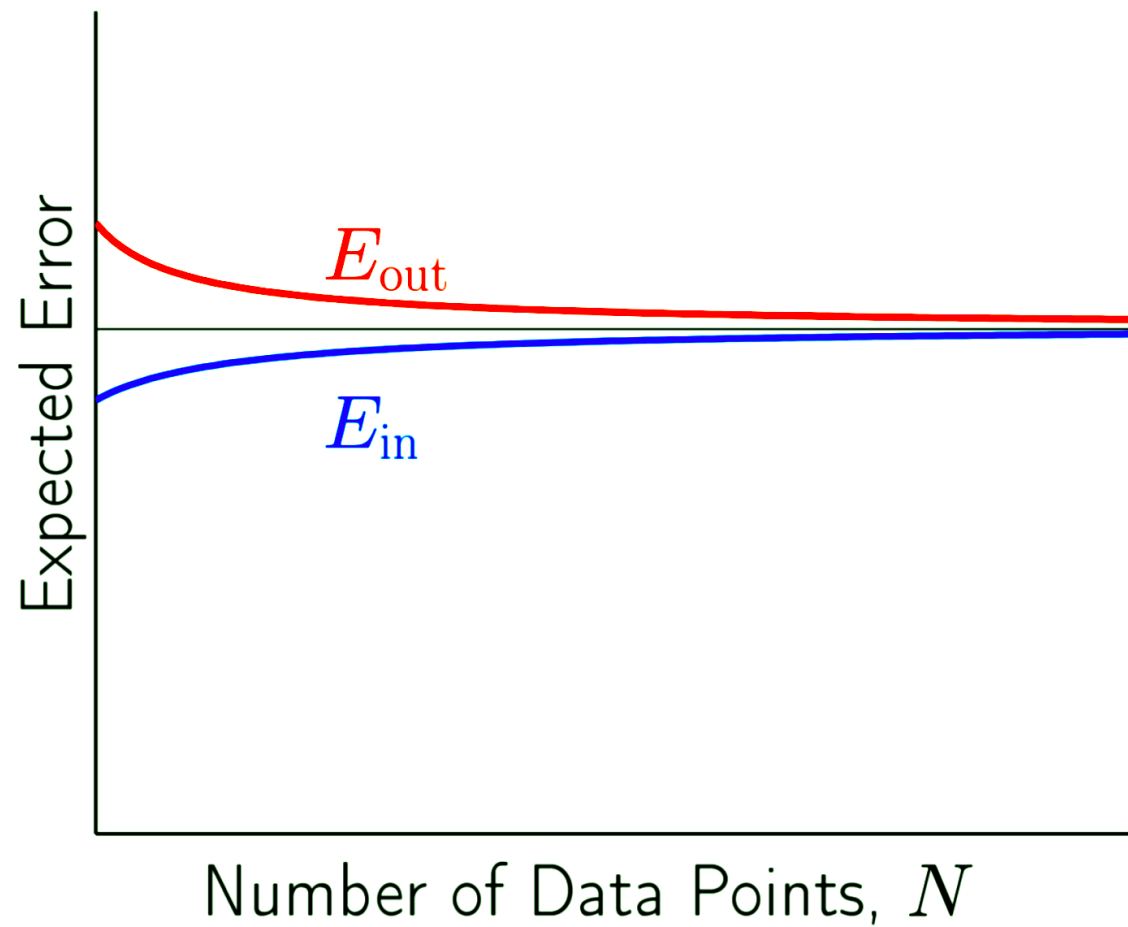
Data set $\mathcal{D}$ of size N

Expected out-of-sample error $\mathbb{E}_{\mathcal{D}}[E_{out}(g^{(\mathcal{D})})]$

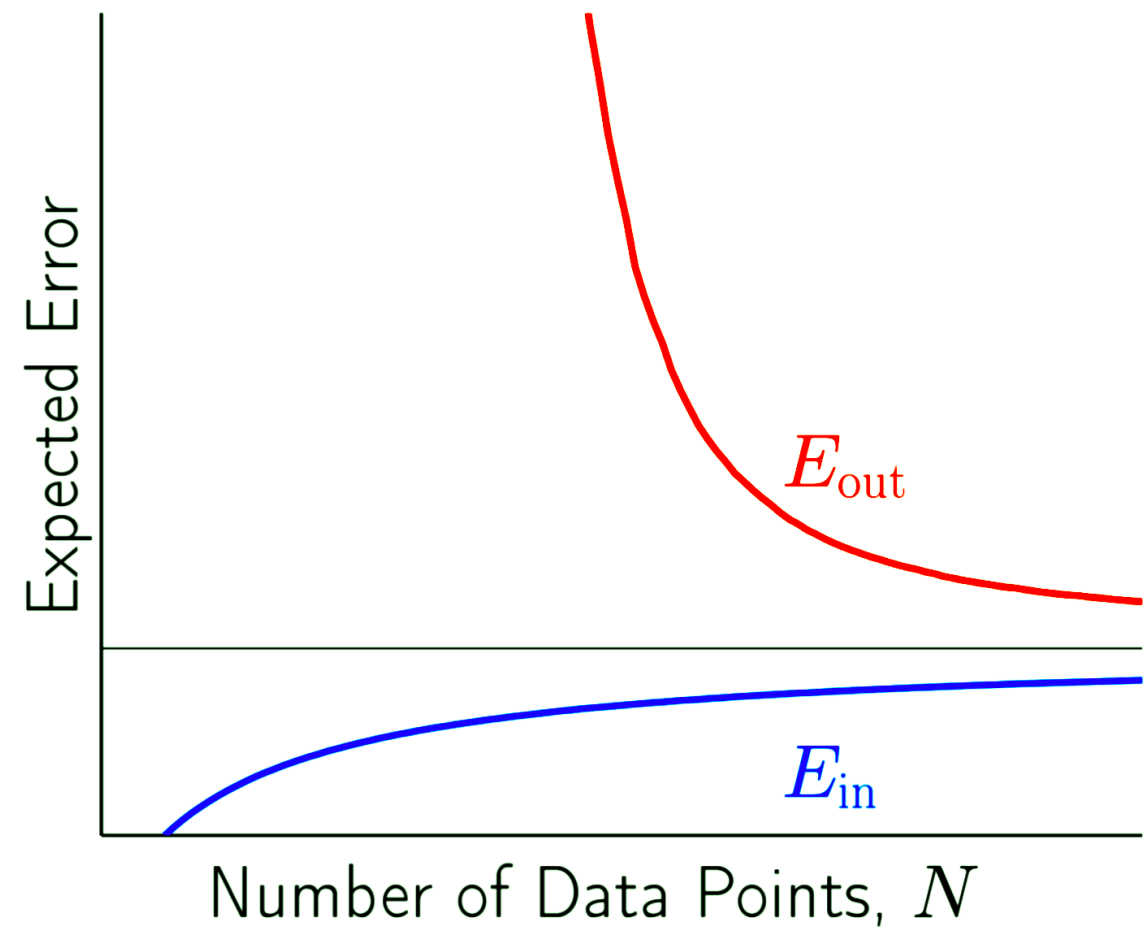
Expected in-sample error $\mathbb{E}_{\mathcal{D}}[E_{in}(g^{(\mathcal{D})})]$

How do they vary with N ?

Learning curves



Simple Model



Complex Model

VC versus bias-variance

